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Interactive system and device with optical sensing function

Abstract

Glasses with gesture recognition function include a glasses frame and a gesture recognition system. The gesture recognition system is disposed on the glasses frame and configured to detect hand gestures in front of the glasses thereby generating a control command. The gesture recognition system transmits the control command to an electronic device to correspondingly control the electronic device.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of U.S. patent application Ser. No. 17/194,585 filed on, Mar. 8, 2021, which is a continuation application of U.S. patent application Ser. No. 16/780,079 filed on, Feb. 3, 2020, which is a continuation application of U.S. patent application Ser. No. 16/196,898 filed on, Nov. 20, 2018, which is a continuation application of U.S. patent application Ser. No. 15/854,498 filed on, Dec. 26, 2017, which is a continuation application of U.S. patent application Ser. No. 13/929,061 filed on, Jun. 27, 2013, which claims the priority benefit of Taiwan Patent Application Serial Number 101124573, filed on Jul. 6, 2012, the full disclosures of which are incorporated herein by reference. (2) To the extent any amendments, characterizations, or other assertions previously made (in this or in any related patent applications or patents, including any parent, sibling, or child) with respect to any art, prior or otherwise, could be construed as a disclaimer of any subject matter supported by the present disclosure of this application, Applicant hereby rescinds and retracts such disclaimer. Applicant also respectfully submits that any prior art previously considered in any related patent applications or patents, including any parent, sibling, or child, may need to be re-visited.

BACKGROUND

1. Field of the Disclosure

(1) This disclosure generally relates to eyeglasses and, more particularly, to glasses with gesture recognition function.

2. Description of the Related Art

(2) With the vigorous development of the smartphone technology, the sensitivity and smoothness of the touch panel are also improved for being operated and used on the users' cell phones. More and more control gestures have been provided such as moving the thumb closer to or farther away the index finger to perform zoom-in and zoom-out functions, or moving the finger rightward to perform a page-turning command. However, the gesture recognition methods mentioned above are all performed through a touch panel.

(3) In the present day, the optical gesture recognition has been developed quickly due to the vigorous development of the multimedia technology. For example, a user can control a smart TV to switch channel or adjust volume by using hand gestures. However, current gesture recognition devices are integrated with the electronic device and the gesture recognition is performed by means of optical sensing such that the operating distance is limited, e.g. the technology disclosed in the U.S. Pat. No. 5,594,469.

SUMMARY

(4) The present disclosure provides a gesture recognition system that has a wireless transmission unit configured to transmit the control command to an electronic device at a distance therefrom.

(5) The present disclosure further provides a device that includes the above gesture recognition system so as to control the electronic device at a distance therefrom.

(6) Other objects and advantages of the present disclosure will become more apparent from the technical features of the present disclosure.

(7) The present disclosure provides a device including a frame and an optical sensing system. The frame includes a left frame, a right frame and a connecting part connecting between the left frame and the right frame, wherein the left frame, the right frame and the connecting part are integrally formed. The optical sensing system includes an illumination device and an image sensor. The image sensor is configured to capture image data, wherein the connecting part connects the right and left frames, and has an upper edge and a lower edge, and the illumination device and the image sensor are both disposed inside a front surface of the connecting part between the upper edge and the lower edge, and the illumination device and the image sensor are adjacent to each other inside the front surface of.

(8) The present disclosure further provides an interactive system including a monitor and a device. The device includes a frame and an optical sensing system. The frame includes a left frame, a right frame and a connecting part connecting between the left frame and the right frame, wherein the left frame, the right frame and the connecting part are integrally formed. The optical sensing system includes an illumination device, an image sensor, a processor and a transmission unit. The image sensor is configured to capture images, wherein the connecting part connects the right and left frames, and has an upper edge and a lower edge, and the illumination device and the image sensor are both disposed inside a front surface of the connecting part between the upper edge and the lower edge, and the illumination device and the image sensor are adjacent to each other inside the front surface. The processor is configured to recognize motion track of an object according to the captured images to generate a control command. The transmission unit is configured to transmit the control command to the monitor to correspondingly control information shown thereon.

(9) The present disclosure further provides an interactive system including a portable device and an electronic device. The portable device includes a frame and an optical sensing system. The frame includes a left frame, a right frame and a connecting part connecting between the left frame and the right frame, wherein the left frame, the right frame and the connecting part are integrally formed. The optical sensing system includes an illumination device, an image sensor and a transmission unit. The image sensor is configured to capture images, wherein the connecting part connects the right and left frames, and has an upper edge and a lower edge, and the illumination device and the

image sensor are both disposed inside a front surface of the connecting part between the upper edge and the lower edge, and the illumination device and the image sensor are adjacent to each other inside the front surface. The transmission unit is configured to transmit the image data captured by the image sensor. The electronic device includes a processor configured to process the image data to generate a control command.

(10) As mentioned above, the gesture recognition system of the present disclosure includes a wireless transmission unit so that the gesture recognition system has an operating distance. As the gesture recognition system may be disposed on the glasses or other portable devices, e.g. a Bluetooth earphone, a cell phone or a tablet computer, it may be placed close to the user all the time. The control command may be transmitted to the electronic device to be controlled through the wireless communication unit of the gesture recognition system to significantly extend the operating distance compared to traditional systems. In addition, the gesture recognition system may be disposed at different positions of the glasses so as to improve the gesture operation. For example, the gesture recognition system may be disposed at the connecting part or the periphery of edges so that the gesture operation may have a lower chance to block the line of sight. Furthermore, if the pair of glasses itself includes the display unit for displaying images, the gesture recognition system may not include the wireless transmission unit; i.e. the control command generated by the processor may be directly transmitted to the display unit to correspondingly control the content shown on the glasses.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other objects, advantages, and novel features of the present disclosure will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

(2) FIG. 1 shows a schematic diagram of the glasses with gesture recognition function according to an embodiment of the present disclosure.

(3) FIG. 2 shows a schematic block diagram of the gesture recognition system shown in FIG. 1.

(4) FIG. 3 shows a schematic diagram of the glasses with partially integrated structure.

(5) FIG. 4 shows a schematic diagram of the gesture recognition system in which the processor is integrated with an electronic device.

(6) FIGS. 5 and 6 show schematic diagrams of the glasses and gesture recognition system thereof according to another embodiment of the present disclosure.

(7) FIGS. 7 and 8 show schematic diagrams of the glasses and gesture recognition system thereof according to an alternative embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENT

(8) The above or other technical contents, characteristics and effects according to the present disclosure will become more apparent from the following detailed description of a preferred embodiment in conjunction with the accompanying drawings. It is to be understood that terms of direction used herein, such as upward, downward, leftward, rightward, forward and backward, are only used for reference but not used to limit the present disclosure.

(9) FIG. 1 shows a schematic diagram of the glasses with gesture recognition function according to an embodiment of the present disclosure. Referring to FIG. 1, the glasses with gesture recognition function **100** of this embodiment includes a glasses frame **110**, wherein a left lens **122** and a right lens **124** may be embedded in the glasses frame **110**. Specifically speaking, the glasses frame **110** includes a left lens frame **112**, a right lens frame **114**, a connecting part **116**, a left temple arm **118a** and a right temple arm **118b**. The connecting part **116** is configured to connect the left lens frame **112** and the right lens frame **114**. The left temple arm **118a** is connected to the left lens frame **112**

and the right temple arm **118b** is connected to the right lens frame **114**, wherein the left and right temple arms **118a**, **118b** are used to support and fix the glasses **100** on two sides of a user's head. (10) In one embodiment, the glasses frame **110** may have an integrated structure; i.e. the left lens frame **112**, the right lens frame **114**, the connecting part **116** and the left and right temple arms **118a**, **118b** are formed integrally. The glasses frame **110** of this embodiment is described with an example having a partially integrated structure; i.e. only the left lens frame **112**, the right lens frame **114** and the connecting part **116** are formed integrally as shown by the glasses **100** in FIG. 3, but the present disclosure is not limited thereto.

(11) Referring to FIG. 1, the glasses **110** of this embodiment further include a gesture recognition system **130**, wherein the gesture recognition system **130** may be disposed on or in the glasses frame **110**. For example, the gesture recognition system **130** may be disposed on the connecting part **116** of the glasses frame **110**, wherein an example of the gesture recognition system **130** in this embodiment may be embedded in the connecting part **116**. In another embodiment, a part of the gesture recognition system **130** may be embedded in the lens frames **112**, **114** or the temple arms **118a**, **118b** and the rest part thereof is exposed outside the lens frames **112**, **114** or the temple arms **118a**, **118b**. In this manner, the hand waving of a user may not be performed in front of the user and the hand waving may be performed at the left-front or right-front of the user in order not to block the line of sight. In another embodiment, the gesture recognition system **130** may also be hung on the glasses **100** such that users with myopia may directly hang the gesture recognition system **130** on their glasses without using an additional pair of glasses.

(12) In this embodiment, the gesture recognition system **130** may include an image sensor **132** and a processor **134**, wherein the image sensor **132** is configured to successively capture images that contain a motion track of the user's hand, and the processor **134** is configured to receive the images and generate a control command C1 according to the motion track of the user's hand as shown in FIGS. 1 and 2, wherein FIG. 2 is a schematic block diagram of the gesture recognition system **130** of FIG. 1.

(13) In one embodiment, the gesture recognition system **130** may include a transmission unit **136** configured to transmit the control command C1 to an electronic device **200** to correspondingly control the electronic device **200**, e.g. controlling the ON/OFF of the electronic device **200**, zoom-in or zoom-out of the screen information shown on the electronic device **200**, the cursor shown on the screen of the electronic device **200**, or the page-turning command of the screen information shown on the electronic device **200**. In other words, the control command C1 may be an ON/OFF command, a zoom-in or zoom-out command, a command turning to a previous page or a next page, a volume command or a channel-selection command.

(14) In this embodiment, the electronic device **200** may be a display device (e.g. a TV or a monitor), a portable electronic device (e.g. a cell phone or a tablet computer), home appliance or a vehicle electronic device. More particularly, the electronic device **200** preferably includes a receiving unit **210** configured to receive the control command C1. In other words, the glasses **100** of this embodiment may wirelessly transmit the control command C1 to the electronic device **200** to accordingly control the electronic device **200**. In this manner, it is able to provide control on the electronic device **200** at a distance therefrom without being limited by a far distance between the user and the electronic device to avoid the condition that the gesture recognition system is unable to detect gestures of the user's hand. It should be mentioned that, in some cases, the user may not want to use the wireless communication due to the electromagnetic radiation and the gesture recognition unit **130** of the glasses **100** may be connected to the electronic device **200** via a cable in another embodiment.

(15) In addition, in order to improve the performance of recognizing the hand motion, the gesture recognition system **130** may further include an illumination device **138**, wherein the illumination device **138** is configured to provide a light beam L1 and the image sensor **132** is configured to receive reflection of the light beam L1 reflected from the hand motion of the user and to generate a

plurality of images. Similarly, the processor **134** is configured to receive the images and generate the control command C1 mentioned above according to the motion track of the user's hand. In this embodiment, the illumination device **138** may be one capable of providing infrared light, whereas the image sensor **132** may be one capable of sensing infrared light, wherein as there is generally no object between the hand motion of the user and the glasses **100**, the gesture recognition system **130** on the glasses **100** is able to recognize the motion track of the user's hand correctly.

(16) In one embodiment, the glasses **100** may further include a switch device **140**, wherein the switch device **140** may be disposed on the glasses frame **110** as shown in FIG. 1. Specifically speaking, the switch device **140** may be configured to turn on or turn off the gesture recognition system **130** so as to effectively reduce the power consumption as well as prevent recognition error.

(17) In addition, the glasses **100** of this embodiment may or may not be embedded with the lens **122**, **124** according to the preference of the user. In one embodiment, the left and right lens **122**, **124** in the glasses frame **110** of the glasses **100** may be corrective lenses, plain glass lenses or lens capable of displaying images, wherein the lens capable of displaying images may include OLED display panel having high transparency as shown by the glasses **100a** and gesture recognition system **130a** thereof in FIGS. 5 and 6. Specifically speaking, as the lens **122a**, **124b** of the glasses **110a** may directly display images, the wireless transmission unit **136** mentioned above may not be implemented in the gesture recognition system **130a** and the processor **134** may directly transmit the control command C1 to the lens **122a**, **124a** to directly control the content shown on the lens **122a**, **124a**.

(18) In another embodiment, the content shown on the lens may also be formed by projection; i.e. the glasses **100b** may further include a projection device **150**, wherein the projection device **150** may directly project images onto the lens **122**, **124** as shown by the glasses **100a** and gesture recognition system **130b** thereof in FIGS. 7 and 8. Similarly, as the lens **122**, **124** of the glasses **100b** may directly show images through projection, the wireless transmission unit **136** may not be implemented in the gesture recognition system **130a** and the processor **134** may directly transmit the control command C1 to the projection device **150** to directly control the content projected on the lens **122**, **124**.

(19) It should be mentioned that the gesture recognition systems **130**, **130a** and **130b** generate the aforementioned control command by using their own processor **134** to process the gesture images received. In one embodiment, the processor **134** may be integrated with the electronic device **200**; i.e. the gesture recognition system **130c** may utilize a control unit **131** to transmit the captured image data I1 to the electronic device **200** through the transmission unit **136** and the processor **134** of the electronic device **200** processes the image data to generate the control command C1 mentioned above as shown in FIG. 4, wherein FIG. 4 shows a schematic diagram of the gesture recognition system in which the processor is integrated with the electronic device **200**.

(20) As mentioned above, the gesture recognition system and the glasses with gesture recognition function according to the embodiment of the present disclosure have at least following advantages. First, the gesture recognition system has a wireless transmission unit so that the gesture recognition system has an operating distance from the devices to be controlled. As the gesture recognition system may be disposed on the glasses or other portable devices, e.g. a Bluetooth earphone, a cell phone or a tablet computer, it may be placed close to the user all the time. The control command may be transmitted to the electronic device to be controlled through the wireless communication unit of the gesture recognition system to significantly extend the operating distance compared to traditional systems.

(21) In addition, the gesture recognition system may be located at different positions of the glasses so as to improve the gesture operation. For example, the gesture recognition system may be disposed at the connecting part or the periphery of edges so that the gesture operation may have a lower chance to block the line of sight. Furthermore, if the glasses have the display unit for displaying images, the gesture recognition system may not include the wireless transmission unit;

i.e. the control command generated by the processor may be directly transmitted to the display unit to accordingly control the content shown on the glasses.

(22) Although the disclosure has been explained in relation to its preferred embodiment, it is not used to limit the disclosure. It is to be understood that many other possible modifications and variations can be made by those skilled in the art without departing from the spirit and scope of the disclosure as hereinafter claimed. Furthermore, any embodiment or claim of the present invention is not necessary to achieve all objects, advantages, and novel features disclosed herein. Meanwhile, the summary and title are only for searching of patent documents but not to limit the disclosure.

Claims

1. A device, comprising: a frame comprising a left frame, a right frame and a connecting part connecting between the left frame and the right frame, wherein the left frame, the right frame and the connecting part are integrally formed, and a width of the connecting part increases from a center thereof to two sides that connect to the left frame and the right frame respectively; and an optical sensing system, comprising: an illumination device configured to provide light toward front of the optical sensing system; and an image sensor configured to capture image data, wherein the connecting part connects the right and left frames at the two sides, and has an upper edge and a lower edge, and the illumination device and the image sensor are both disposed inside a front surface of the connecting part between the upper edge and the lower edge, and the illumination device and the image sensor are adjacent to each other inside the front surface.
2. The device as claimed in claim 1, wherein the connecting part is narrower than the right frame and the left frame.
3. The device as claimed in claim 1, wherein the illumination device has a same height with the image sensor inside the front surface of the connecting part.
4. The device as claimed in claim 1, wherein the upper edge of the connecting part extends between upper edges of the right frame and the left frame, and the lower edge of the connecting part extends between lower edges of the right frame and the left frame.
5. The device as claimed in claim 1, wherein the illumination device is configured to emit infrared light.
6. The device as claimed in claim 1, wherein the frame further comprises a left temple arm and a right temple arm respectively connected to the left frame and the right frame.
7. The device as claimed in claim 1, further comprising a switch device arranged on the frame, and the switch device is configured to turn on or turn off the optical sensing system.
8. The device as claimed in claim 1, further comprising a transmission unit configured to wirelessly transmit a control command.
9. The device as claimed in claim 1, further comprising a transmission unit configured to transmit the image data captured by the image sensor.
10. An interactive system, comprising: a monitor; and a device, comprising: a frame comprising a left frame, a right frame and a connecting part connecting between the left frame and the right frame, wherein the left frame, the right frame and the connecting part are integrally formed, and a width of the connecting part increases from a center thereof to two sides that connect to the left frame and the right frame respectively; and an optical sensing system, comprising: an illumination device configured to provide light toward front of the optical sensing system; an image sensor configured to capture images, wherein the connecting part connects the right and left frames at the two sides, and has an upper edge and a lower edge, and the illumination device and the image sensor are both disposed inside a front surface of the connecting part between the upper edge and the lower edge, and the illumination device and the image sensor are adjacent to each other inside the front surface; a processor configured to recognize motion track of an object according to the captured images to generate a control command; and a transmission unit configured to transmit the

control command to the monitor to correspondingly control information shown thereon.

11. The interactive system as claimed in claim 10, wherein the control command comprises an ON/OFF command, a zooming command, a page-turning command, a volume command and a channel-selection command.

12. The interactive system as claimed in claim 10, wherein the motion track comprises an upward track, a downward track, a leftward track, a rightward track, a forward track, a backward track and a rotation track of the object.

13. The interactive system as claimed in claim 10, wherein the connecting part is narrower than the right frame and the left frame.

14. The interactive system claimed in claim 10, wherein the illumination device has a same height with the image sensor inside the front surface of the connecting part.

15. The interactive system as claimed in claim 10, wherein the upper edge of the connecting part extends between upper edges of the right frame and the left frame, and the lower edge of the connecting part extends between lower edges of the right frame and the left frame.

16. The interactive system as claimed in claim 10, wherein the device further comprises a left temple arm and a right temple arm respectively connected to the left frame and the right frame.

17. The interactive system as claimed in claim 10, wherein the device further comprises a switch device arranged on the frame, and the switch device is configured to turn on or turn off the optical sensing system.

18. An interactive system, comprising: a portable device, comprising: a frame comprising a left frame, a right frame and a connecting part connecting between the left frame and the right frame, wherein the left frame, the right frame and the connecting part are integrally formed, and a width of the connecting part increases from a center thereof to two sides that connect to the left frame and the right frame respectively; and an optical sensing system, comprising: an illumination device configured to provide light toward front of the optical sensing system; an image sensor configured to capture image data, wherein the connecting part connects the right and left frames at the two sides, and has an upper edge and a lower edge, and the illumination device and the image sensor are both disposed inside a front surface of the connecting part between the upper edge and the lower edge, and the illumination device and the image sensor are adjacent to each other inside the front surface; and a transmission unit configured to transmit the image data captured by the image sensor; and an electronic device comprising a processor configured to process the image data to generate a control command.

19. The interactive system as claimed in claim 18, wherein the electronic device comprises a display.

20. The interactive system as claimed in claim 18, wherein the upper edge of the connecting part extends between upper edges of the right frame and the left frame, and the lower edge of the connecting part extends between lower edges of the right frame and the left frame.
